What Is a Good Model?

Provost & Fawcett, Data Science for Business, chapters 7–8

AI in Business · Week 3

Last week's cliffhanger, and today's plan

74%

of the 200 customers ranked highest by our hand-made score really churned, against 26.5% on average.

But should we call them? How many?

1

What is a good model?

Ch. 7: accuracy, confusion matrix, expected value, baselines

45 min

2

Seeing model performance

Ch. 8: ranking, profit, lift and ROC curves

40 min

1. What is a good model?

Chapter 7: decision-analytic thinking

Accuracy can look good and mean nothing!

99.9%

accuracy from always predicting "no" when classes are 999 : 1

90%

accuracy from "nobody churns" when 10% of customers churn (MegaTelCo)

73.5%

accuracy from "nobody churns" on our Telco data. Useless, yet it sounds fine

Always compare with the base rate. Accuracy = correct decisions / all decisions. With unbalanced classes, predicting the majority class already scores high.

The confusion matrix: four kinds of decisions

Predicted Class	Actual Class: Positive	Actual Class: Negative
Churn	True positive	False positive
Not Churn	False negative	True negative

In churn terms (Jupyter Notebook)

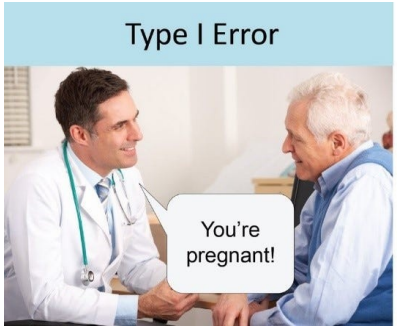
True positive: we predicted churn, and they left

False positive: we predicted churn, but they stayed (Type I Error)

False negative: we said they would stay, but they left (Type II Error)

True negative: we said they would stay, and they did

Note: In statistics and ML, "positive" means worth our attention (eg. death is positive, living is 0/negative)
In business cases, Positive means churn, fraud, disease, not necessary "good" in sociological terms!





Same accuracy, very different models

Model A



Model B



Balanced test set (50 : 50)	**A: 80%**	**B: 80%**
Real population (10% churn)	**A: 64%**	**B: 96%**

Which is better? We cannot say without costs, and context!

Example from ch. 7: 1,000 customers. A catches every churner but raises many false alarms; B never raises a false alarm but misses most churners.

Errors do not cost the same (Cost-Benefit)



Medical test

False positive

Told she has cancer, but does not: more tests, stress, expense.

False negative

Told she is healthy, but has cancer: missed early treatment.



Churn

False positive

Retention offer to someone who stays anyway: we pay for nothing.

False negative

No offer to someone who leaves: we lose a customer.

The general principle: When we are applying data science to an actual application it is vital to return to the question: what is important in the application? What is the goal? Are we assessing the results of data mining appropriately given the actual goal?

Expected value: the framework for the whole course

$$EV = p(o_1) \cdot v(o_1) + p(o_2) \cdot v(o_2) + p(o_3) \cdot v(o_3) + \dots$$

Probability of each outcome × its value, added up: a dot product, as in week 2.



Structure

Which outcomes are possible? Lay out the decision.



Probabilities

Usually estimated from data, by a model.



Values

Usually from the business: costs, margins, experts.

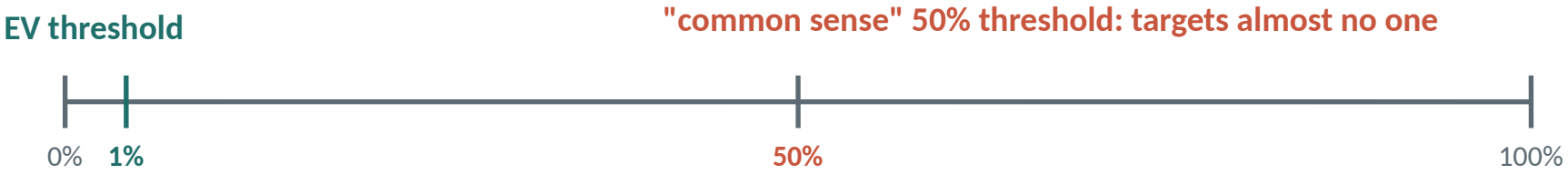
Should we target this customer? A threshold from EV

A product sells for $200 and costs $100 to make. Mailing an offer costs $1.

If she buys
+$99

If she does not
−$1

Target if $p \times 99 - (1 - p) \times 1 > 0$
→ p > 1%



Response rates are often 1–2%. With a 50% cut-off, nobody is ever a "likely responder".

Bottom Line: Understanding Problem is Crucial → Domain Expertise + Machine Learning!

Activity 1: Who gets the offer?



In pairs · 8 minutes · calculator allowed

An offer costs €2 to send. A customer who responds brings €38 profit after that cost; one who does not respond costs us the €2.

Lali
p(respond) = 2%

Fiku
p(respond) = 4%

Noci
p(respond) = 8%

Rroku
p(respond) = 30%

1. Compute the expected value of targeting each customer.
2. What is the break-even probability?

A whole model's value: confusion matrix × benefits

Test-set counts (110 consumers)

		Actual Class: Positive	Actual Class: Negative
	Y	56	7
	N	5	42

×

Benefit of each decision ($)

		Actual Class: Positive	Actual Class: Negative
	Y	+99	−1
	N	0	0

=

≈ $50
expected profit per consumer

$$56/110 \times \$99 \;+\; 7/110 \times (-\$1) \;+\; 0 \;+\; 0$$

Divide each count by the total to get a probability, multiply by its value, add up. Now models can be compared in money, on the same population.

A false negative: Predicted to be false, nothing was offered, no spending = 0
A true negative: Predicted negative, is negative, no offer, no buy = 0

Always beat a sensible baseline, not just random

Majority class

Predict the most common class. 94% accuracy is nothing if 94% are negatives.

Simple rules

Weather: "same as today" and "historical average" are hard to beat.

One feature

A decision stump: use trees to select one informative variable. Often surprisingly good.

Received wisdom

What experts already do: e.g. go after the biggest past customers.

Our baseline in the lab: last week's hand-made risk score. Does a real model beat it?

2. Seeing model performance

Chapter 8: profit, lift and ROC curves

Rank customers, then choose where to cut

"A different strategy for making decisions is to rank a set of cases by these scores, and then take actions on the cases at the top of the ranked list".

Score	Really churned?	
0.91	yes (p)	
0.84	yes (p)	
0.77	no (n)	
		Cut A: top 3
0.70	yes (p)	
0.62	no (n)	
0.55	yes (p)	
		Cut B: top 6
0.41	no (n)	
0.33	no (n)	
0.24	yes (p)	
0.12	no (n)	

Each cut gives a different confusion matrix

Cut A

		Actual Class: Positive	Actual Class: Negative
Y		2	1
N		3	4

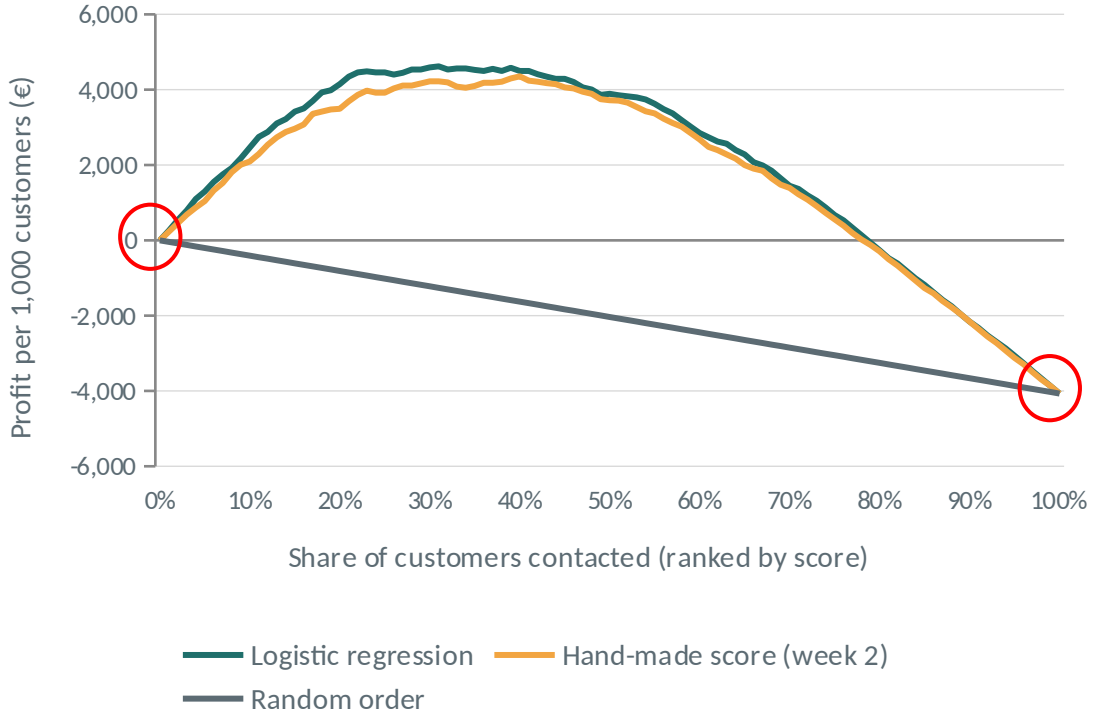
Cut B

		Actual Class: Positive	Actual Class: Negative
Y		4	2
N		1	3

Why rank instead of classify?

- A fixed budget: call the top N
- Scores may rank well even if they are not exact probabilities
- Costs may be uncertain: see the curves next

Profit curve: how far down the list should we go?



Assumptions

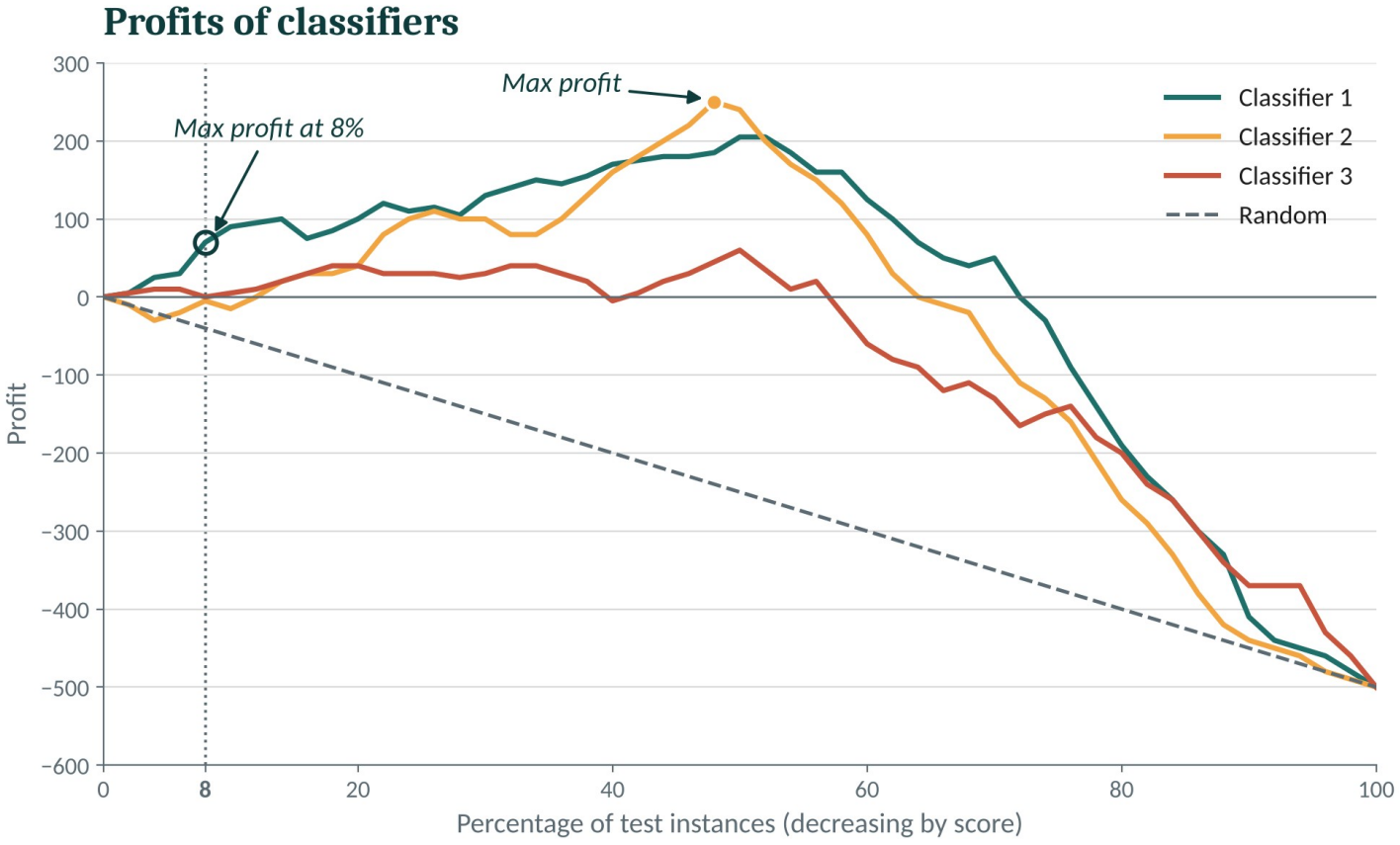
A call plus discount costs €20. A contacted churner is kept 1 time in 4, worth €240 → a true positive is worth +€40; a false positive −€20.

Best: contact the top 31%
€4,619 per 1,000 customers

Contact everyone
−€4,070 per 1,000 customers

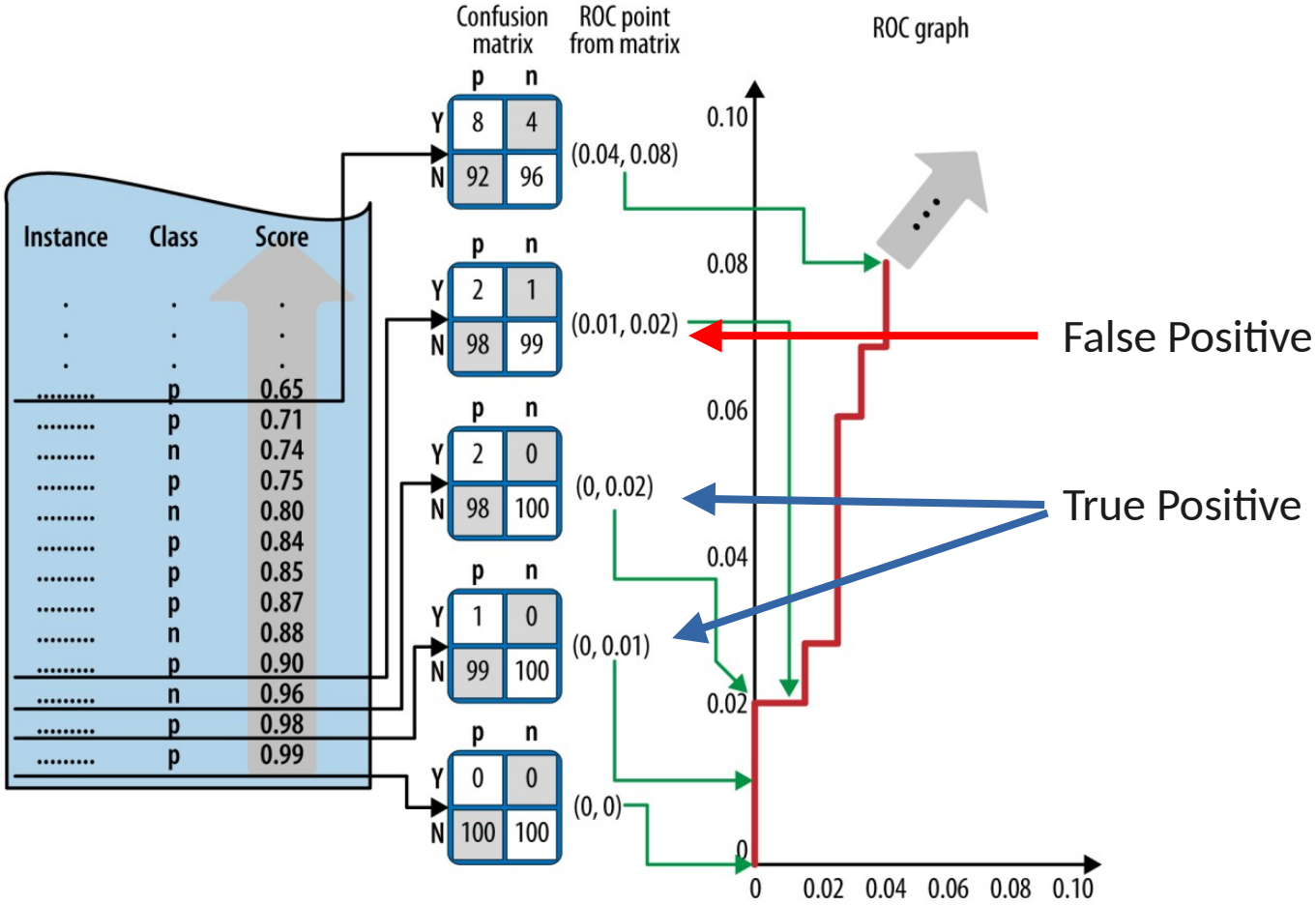
Telco test set (2,113 customers, churn 26.5%). Assumptions are illustrative; you will change them in the lab.

Profit curve: What about budgetary constraints?

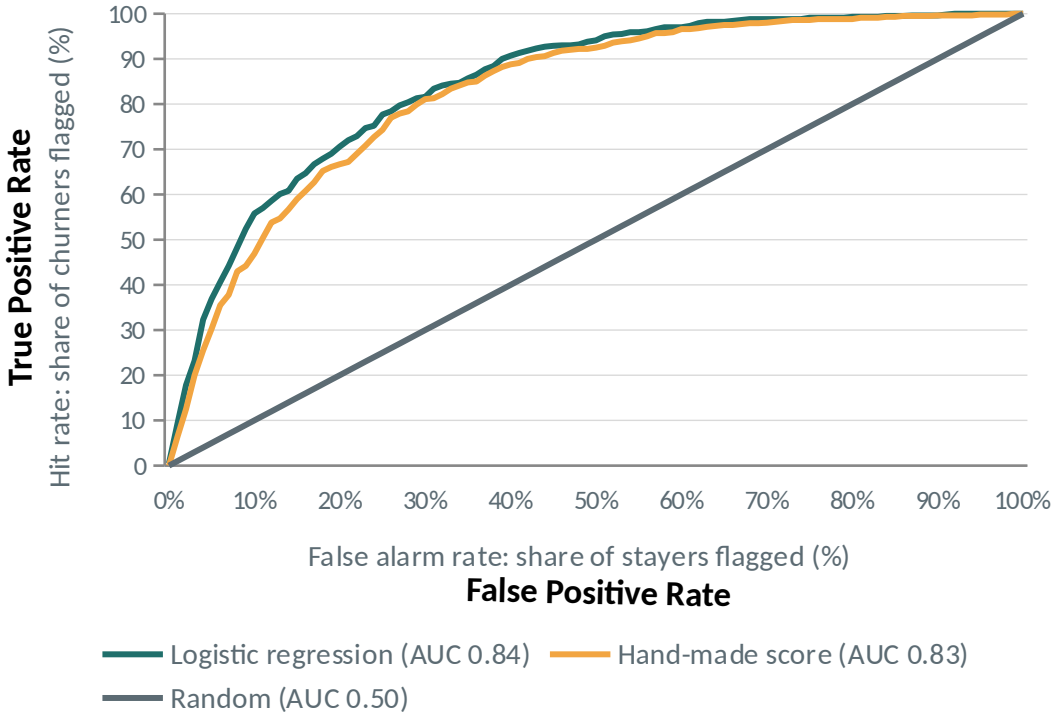


- 100k customers
- $40,000 for the marketing campaign
- 4000/5 = 800 customers

Step-by-step building of the ROC Curve



ROC and AUC: ranking quality apart from costs



- Each point is one cut-off: hit rate vs. false alarm rate
- Top-left corner = perfect. Diagonal = guessing
- Does not change with costs or the share of churners

AUC

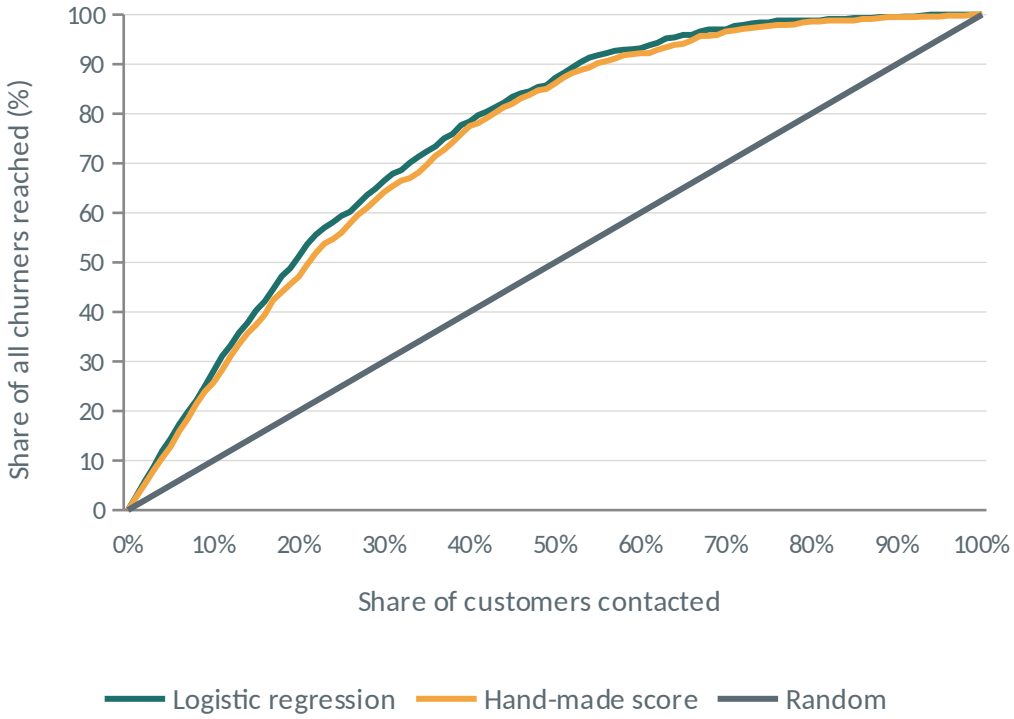
Area under the curve, 0.5 to 1. The chance that a random churner is ranked above a random stayer.

ROC

ROC graph depicts relative trade-offs that a classifier makes between benefits (true positives) and costs (false pos- itives)

Cumulative response and lift: no costs needed

Cumulative response curves plot the hit rate (tp rate; y axis), i.e., the percentage of positives correctly classified, as a function of the percentage of the population that is targeted (x axis)



2.8× lift

The top 10% of the list holds 28% of all churners. Random would hold 10%.

- Lift = model's hit rate / random hit rate at the same point
- Easy to explain to managers
- Assumes the test set has the real share of churners

Which chart when?



Profit curve

Use when: Costs, benefits and class mix are known and stable

Best for managers: shows money



Cumulative response / lift

Use when: Class mix known, costs uncertain

Easy to explain: "top 10% holds 28% of churners"



ROC and AUC

Use when: Costs and class mix uncertain or changing

For analysts comparing models

Rule of thumb: show managers profit or lift; keep ROC and AUC for comparing models.

The book's churn case: accuracy points the wrong way

Model	Accuracy on training data	Accuracy (cross-validated)	AUC (cross-validated)
Classification tree	95%	91.8%	0.614
Logistic regression	93%	93.0%	0.574
k-nearest neighbour	**100%**	93.0%	**0.537**
Naive Bayes	76%	**76.5%**	**0.632**

- 100% on training data = memorising, not learning (overfitting).
- k-NN's 93% just copies the base rate: it almost never predicts churn.
- Naive Bayes has the lowest accuracy but ranks churners best (highest AUC).
- Cost-benefit matrix may help to choose bwtween LR and NB

KDD Cup 2009, Orange (France Telecom): about 47,000 customers, 7% churn. Ch. 8, tables 8-1 and 8-2.

Activity 2: Read the profit curve

1. No budget limit: which ranking would you use, and how far down the list would you go?
2. Budget for only 10% of customers: which ranking, and what profit do you expect?
3. Marketing proposes calling every customer "to be safe". What do you answer?
4. The offer turns out to cost €30, not €20. Which way does the best cut-off move, and why?

Five things to remember

1. Accuracy hides what matters: always check the base rate
2. Count each kind of error separately; they cost different amounts
3. Expected value = probabilities from data × values from the business
4. The best threshold comes from costs, not from 50%
5. Pick the chart for your uncertainty: profit, lift or ROC